

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No-6	Team No-12	Project Title - 207 College Canteen Token System
Team Lead Roll Number	2520090055	
Team Lead Name	Sai Saketh Vutharavalli	
Team Member Roll Number	2520030546	
Team Member Name	BURLLA CHENNA MALLESWAR REDDY	

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

In today's fast-paced academic environment, long lines and manual order management in college canteens waste time for students and create inefficiencies for canteen staff. The College Canteen Token System is a web application that aims to digitize and simplify the entire meal pre-ordering process within a college campus.

This system allows students to browse the daily menu, apply dietary filters (vegetarian or non-vegetarian), place orders within set daily limits, and complete a mock payment, all through an easy-to-use web interface. After placing an order, the system generates a unique QR-coded token for each student. This token acts as a contactless pickup pass at the canteen counter.

For administrators, canteen staff and managers have a dedicated panel to manage the menu, monitor incoming orders in real time, track daily revenue through a billing dashboard, and send notifications to students about their order status.

The application uses Node.js and Express.js for the backend REST API, while HTML5, CSS3, and JavaScript support the frontend. LocalStorage is used to save data, so there is no need for complicated database setups. QR code generation is handled through the qrcode npm package, and all APIs are tested with Postman.

This project shows how full-stack web development concepts work together, including RESTful API design, modular frontend structure, role-based access control, and real-time order management. It addresses a common issue on campus effectively.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
Menu Display + Dietary Filter	2520090055	
Token Form + Daily Limit	2520030546	
Payment Mock + Token QR Generation	2520090055 & 2520030456	
Billing Dashboard + Admin Panel	2520090055 & 2520030546	



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Order History + Notifications	2520090055 & 2520030546	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

Member 1 —

Responsible for building the student-facing frontend: displaying the daily menu with veg/non-veg filters, implementing the order form with daily limit validation, integrating mock payment flow, and generating a QR-coded token upon successful order placement.

Member 2 —

Responsible for the backend REST API using Node.js and Express.js, implementing order history retrieval, push-style notifications for order status, the billing dashboard showing daily revenue, and the admin panel for menu and order management.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

RuntimeNode.jsBackend FrameworkExpress.jsFrontendHTML5, CSS3, JavaScript (ES6+)API
TestingPostmanQR Code Generationqrcode (npm package)Package ManagernpmData
StorageLocalStorage / JSON fileVersion ControlGit & GitHubCode EditorVS Code

Guide Name and Signature (remarks if any) _____